

Lab 4 - Fuzzing Report

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Task 1: WebApp Fuzzing

Step 1: Setup Environment

1. Install Tools

Commands:

```
# Install ffuf
sudo apt-get install ffuf

# Download and extract SecLists
wget -c https://github.com/danielmiessler/SecLists/archive/master.zip -O
SecList.zip && unzip SecList.zip && rm -f SecList.zip
```

Explanation:

- Installs **ffuf**, the fuzzing tool.
- Downloads the **SecLists** wordlist repository and extracts it.

```
inflating: SecLists-master/Miscellaneous/domains-resolvers.txt
inflating: SecLists-master/Miscellaneous/domains-million-top.txt
inflating: SecLists-master/Miscellaneous/ike-groupid.txt
inflating: SecLists-master/Miscellaneous/lang-english.txt
inflating: SecLists-master/Miscellaneous/lang-french-full.txt
inflating: SecLists-master/Miscellaneous/lang-french-small.txt
inflating: SecLists-master/Miscellaneous/lang-german.txt
inflating: SecLists-master/Miscellaneous/lang-portuguese.txt
inflating: SecLists-master/Miscellaneous/lang-spanish.txt
inflating: SecLists-master/Miscellaneous/pl-large.txt
inflating: SecLists-master/Miscellaneous/schemes.txt
extracting: SecLists-master/Miscellaneous/top-domains-alexa.csv.zip
inflating: SecLists-master/Miscellaneous/top-domains-majestic.csv.zip
inflating: SecLists-master/Miscellaneous/us-cities.txt
inflating: SecLists-master/Miscellaneous/wordlist-skipfish.fuzz.txt
creating: SecLists-master/Passwords/
extracting: SecLists-master/Passwords/500-worst-passwords.txt.bz2
creating: SecLists-master/Passwords/Books/
inflating: SecLists-master/Passwords/Books/greatest_books_of_all_time_lowercase.txt
inflating: SecLists-master/Passwords/Books/greatest_books_of_all_time_original.txt
inflating: SecLists-master/Passwords/Books/greatest_books_of_all_time_uppercase.txt
inflating: SecLists-master/Passwords/Books/greatest_books_of_all_time_with_leet_variations.txt
creating: SecLists-master/Passwords/Common-Credentials/
inflating: SecLists-master/Passwords/Common-Credentials/10-million-password-list-top-100.txt
inflating: SecLists-master/Passwords/Common-Credentials/10-million-password-list-top-1000.txt
inflating: SecLists-master/Passwords/Common-Credentials/10-million-password-list-top-10000.txt
inflating: SecLists-master/Passwords/Common-Credentials/10-million-password-list-top-100000.txt
inflating: SecLists-master/Passwords/Common-Credentials/10-million-password-list-top-1000000.txt
inflating: SecLists-master/Passwords/Common-Credentials/100k-most-used-passwords-NCSC.txt
inflating: SecLists-master/Passwords/Common-Credentials/10k-most-common.txt
inflating: SecLists-master/Passwords/Common-Credentials/1990-2020.txt
inflating: SecLists-master/Passwords/Common-Credentials/2020-200-most-used-passwords.txt
inflating: SecLists-master/Passwords/Common-Credentials/2023-200-most-used-passwords.txt
inflating: SecLists-master/Passwords/Common-Credentials/2024-197-most-used-passwords.txt
inflating: SecLists-master/Passwords/Common-Credentials/500-worst-passwords.txt
creating: SecLists-master/Passwords/Common-Credentials/Language-Specific/
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Cantonese_Pwdb_common-password-list-top-150.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Chinese-common-password-list-top-1000.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Chinese-common-password-list-top-10000.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Chinese-common-password-list-top-100000.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Chinese-common-password-list-top-1000000.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Chinese-common-password-list.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Croatian_Pwdb_common-password-list-top-150.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Danish_Pwdb_common-password-list-top-150.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Dutch_common-password-list.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Estonian_Pwdb_common-password-list-top-150.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/Finnish_Pwdb_common-password-list-top-150.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/French-common-password-list-top-20000.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/French-common-password-list-top-5000.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/French_Pwdb_common-password-list-top-150.txt
inflating: SecLists-master/Passwords/Common-Credentials/Language-Specific/German_Pwdb_common-password-list-top-150.txt
```

```

Inflating: SecLists-master/Web-Shells/Vtiger/modules/vtigerVulnPlugin/actions/gateway.php
Inflating: SecLists-master/Web-Shells/Vtiger/modules/vtigerVulnPlugin/manifest.xml
creating: SecLists-master/Web-Shells/Vtiger/settings/
creating: SecLists-master/Web-Shells/Vtiger/settings/actions/
Inflating: SecLists-master/Web-Shells/Vtiger/settings/actions/Gateway.php
creating: SecLists-master/Web-Shells/WordPress/
Inflating: SecLists-master/Web-Shells/WordPress/bypass-login.php
Inflating: SecLists-master/Web-Shells/WordPress/plugin-shell.php
Inflating: SecLists-master/Web-Shells/backdoor_list.txt
creating: SecLists-master/Web-Shells/Laudanun-1.0/
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/CREDITS
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/GPL
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/README
creating: SecLists-master/Web-Shells/Laudanun-1.0/asp/
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/asp/dns.asp
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/asp/file.asp
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/asp/proxy.asp
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/asp/shell.asp
creating: SecLists-master/Web-Shells/Laudanun-1.0/asp/
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/asp/shell.aspx
creating: SecLists-master/Web-Shells/Laudanun-1.0/cfm/
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/cfm/shell.cfm
creating: SecLists-master/Web-Shells/Laudanun-1.0/jsp/
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/jsp/cnd.war
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/jsp/nakewar.sh
creating: SecLists-master/Web-Shells/Laudanun-1.0/jsp/warfiles/
creating: SecLists-master/Web-Shells/Laudanun-1.0/jsp/warfiles/META-INF/
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/jsp/warfiles/META-INF/MANIFEST.MF
creating: SecLists-master/Web-Shells/Laudanun-1.0/jsp/warfiles/META-INF/
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/jsp/warfiles/META-INF/web.xml
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/jsp/warfiles/cnd.jsp
creating: SecLists-master/Web-Shells/Laudanun-1.0/php/
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/php/dns.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/php/file.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/php/host.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/php/killnc.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/php/php-reverse-shell.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/php/proxy.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/php/shell.php
creating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/Laudanun.php
creating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/templates/
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/templates/dns.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/templates/file.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/templates/host.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/templates/lpcheck.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/templates/killnc.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/templates/php-reverse-shell.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/templates/proxy.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/templates/settings.php
Inflating: SecLists-master/Web-Shells/Laudanun-1.0/wordpress/templates/shell.php
yehia@yehia-Aspire-A315-58: ~/my station/secure development/lab4$ ffuf -r -u http://localhost:80/FUZZ -w ./SecLists-master/Discovery/Web-Content/big.txt

```

2. Start DVWA with Docker

Command:

```
docker run -d -p 127.0.0.1:80:80 vulnerables/web-dvwa
```

Explanation:

- Runs DVWA (Damn Vulnerable Web App) on <http://localhost:80>.

```

yehia@yehia-Aspire-A315-58: ~/my station/secure development/lab4$ docker run -d -p 127.0.0.1:80:80 vulnerables/web-dvwa
d82ce2da878174b114c72044344ef356dc17ed832da09004653327afcf93b9e
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint tender_neumann (35e4cf5bb84dbb714344daf9a668f669fb544039f6c820a4b570894072dfa): Bind for 127.0.0.1:80 failed: port is already allocated

Run 'docker run --help' for more information
yehia@yehia-Aspire-A315-58: ~/my station/secure development/lab4$ wget -c https://github.com/danielmiessler/SecLists/archive/master.zip -O SecList.zip && unzip SecList.zip && rm -f SecList.zip

```

Step 2: Fuzz Endpoints with big.txt

Question: Which endpoints/files from **big.txt** were accessible? Which ones gave interesting error codes (not 404)?

Command:

```
ffuf -r -u http://localhost:80/FUZZ -w ./SecLists-master/Discovery/Web-Content/big.txt
```

Explanation:

- Fuzzes the DVWA server for files/endpoints listed in **big.txt**.
- r** enables following redirects.
- FUZZ** is replaced with entries from the wordlist.

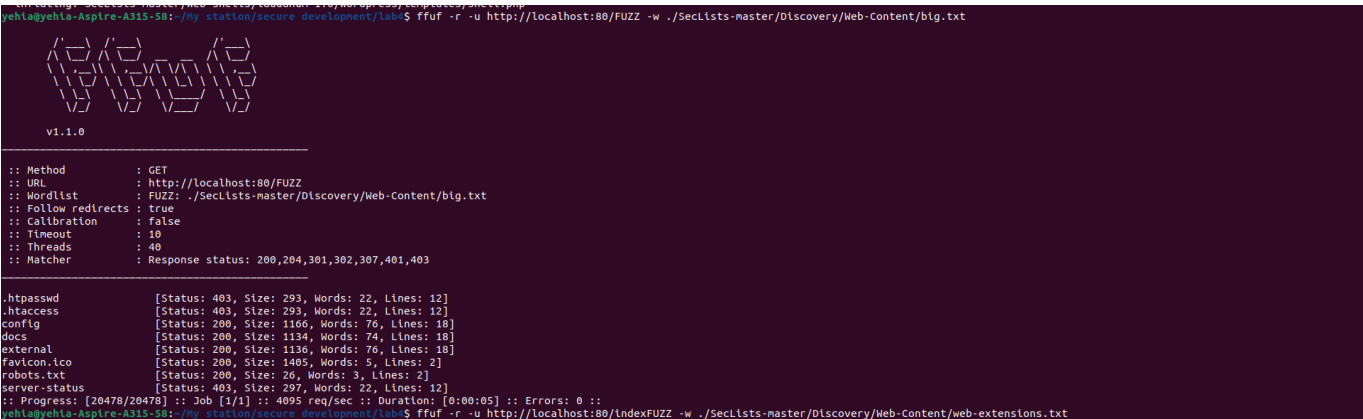
Results:

Endpoint/File	Status Code	Description
.htpasswd	403	Forbidden access
.htaccess	403	Forbidden access
config	200	Accessible
docs	200	Accessible
external	200	Accessible
favicon.ico	200	Accessible
robots.txt	200	Accessible
server-status	403	Forbidden access

Interesting Error Codes:

- 403 Forbidden: .htpasswd, .htaccess, server-status
- 200 OK: config, docs, external, favicon.ico, robots.txt

Screenshot:



Step 3: Fuzz File Extensions for index Page

Question: What file extensions from web-extensions.txt are available for the index page?

Command:

```
ffuf -r -u http://localhost:80/indexFUZZ -w ./SecLists-master/Discovery/Web-Content/web-extensions.txt
```

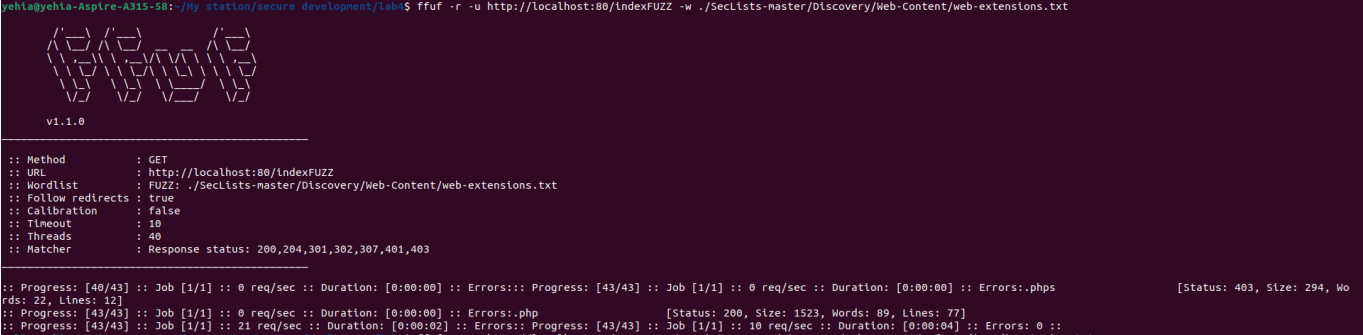
Results:

Extension	Status Code	Description
.php	200	Valid PHP file
.phps	403	Forbidden access

Answer:

- **Valid Extension:** `.php` (returns `200 OK`).
- **Interesting Error:** `.phps` (`403 Forbidden`).

Screenshot:



Step 4: Fuzz Directories with `raft-medium-directories.txt`

Question: Which directories from `raft-medium-directories.txt` are accessible? Which ones gave interesting error codes?

Command:

```
ffuf -r -u http://localhost:80/FUZZ -w ./SecLists-master/Discovery/Web-Content/raft-medium-directories.txt
```

Results:

Directory	Status Code	Description
config	200	Accessible
docs	200	Accessible
external	200	Accessible
server-status	403	Forbidden access

Answer:

- **Accessible Directories:** `config`, `docs`, `external` (`200 OK`).
- **Interesting Error:** `server-status` (`403 Forbidden`).


```
id:0000003,sig:10,src:0000000,time:19902,execs:7445,op:havoc,rep:3
id:0000000,sig:10,src:0000000,time:1837,execs:750,op:havoc,rep:4
id:0000002,sig:10,src:0000000,time:11557,execs:4114,op:havoc,rep:4
id:0000004,sig:10,src:000010,time:39711,execs:15301,op:quick,pos:16
```

2.2 Hangs Found:

```
ls output/default/hangs/
# 2 hang files detected (id:0000000 and id:0000001)
id:0000000,src:0000000,time:4061,execs:1208,op:havoc,rep:3
id:0000001,src:000011,time:41933,execs:15880,op:havoc,rep:6
```

2.3 Key Stats:

```
cat output/default/fuzzer_stats
```

- **Total Executions:** 73,587
- **Crashes Saved:** 5
- **Hangs Saved:** 2
- **Execution Speed:** ~457 execs/sec

Step 3: Reproduce Crashes

3.1 Crash Case 1 (IndexError):

```
python3 main.py < output/default/crashes/id:0000000...
```

Error:

```
Traceback (most recent call last):
  File "/src/main.py", line 23, in <module>
    print(uridecode(sys.stdin.read()))
  File "/src/main.py", line 11, in uridecode
    char_code = (int(a, 16) * 16) + int(b, 16)
ValueError: invalid literal for int() with base 16: 's'
```

Cause:

- Input `%stt` has `%` at position `0` but lacks valid hex characters.

3.2 Crash Case 2 (ValueError):

```
python3 main.py < output/default/crashes/id:000004...
```

Error:

```
ValueError: invalid literal for int() with base 16: '\udc9a'
```

Cause:

The error occurs when:

- Input contains %stt (as seen in crash file id:000000)
- Code tries to convert 's' to hexadecimal with `int(a, 16)`
- Fails because 's' is not a valid hexadecimal digit (only 0-9, a-f, A-F are valid)

Step 4: Fix the Code

Fixed Code:

```
import afl
import sys

def uridecode(s):
    ret = []
    i = 0
    while i < len(s):
        if s[i] == '%':
            # Case 1: Not enough characters after %
            if i + 2 >= len(s):
                ret.append(s[i])
                i += 1
                continue

            a, b = s[i+1], s[i+2]

            # Case 2: Validate hex characters
            hex_chars = set('0123456789abcdefABCDEF')
            if a in hex_chars and b in hex_chars:
                try:
                    char_code = (int(a, 16) * 16) + int(b, 16)
                    ret.append(chr(char_code))
                    i += 3
                except ValueError:
```

```

        # Fallback for unexpected hex conversion errors
        ret.append(s[i])
        i += 1

    else:
        # Preserve invalid % sequences
        ret.append(s[i])
        i += 1

    elif s[i] == '+':
        ret.append(' ')
        i += 1
    else:
        ret.append(s[i])
        i += 1
    return ''.join(ret)

if __name__ == '__main__':
    afl.init()
    print(uridecode(sys.stdin.read()))

```

Step 5: Final Summary

- **Crashes:** Invalid hex characters and incomplete % sequences.
- **Hangs:** Long/malformed inputs.
- **Fix:** Bounds checks, hex validation, error handling.
- **AFL++ Behavior:** Runs indefinitely, coverage-guided, optimized via dictionaries/persistent mode.

Screenshot:

```

yehia@yehia-Aspire-A315-58:~/my_station/secure_development/lab4/task2$ docker run --name afl -ti -v ./src/ aflplusplus/aflplusplus
Unable to find image 'aflplusplus/aflplusplus:latest' locally
latest: Pulling from aflplusplus/aflplusplus
9cb31e2e37ea: Pull complete
ed0c2f843eac: Pull complete
c953e3a86643: Pull complete
8b1d38c8064a: Pull complete
2acc3ff0b072: Pull complete
c65c88fa9805: Pull complete
c637baeec22e: Pull complete
7cab94279aff: Pull complete
918df1abcad6: Pull complete
6d3eff7b5d40: Pull complete
cad47d71df55: Pull complete
862039f7f99f: Pull complete
Digest: sha256:ccd7bffa87fd49ec3f4529003eaf8f75a28fb09e07fb13b14f696480c15196cb
Status: Downloaded newer image for aflplusplus/aflplusplus:latest
[afl++ b0def6d55d05] /aflplusplus # cd /src
[afl++ b0def6d55d05] /src # pip install python-afl
Collecting python-afl
  Downloading python-afl-0.7.3.tar.gz (16 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Building wheels for collected packages: python-afl
  Building wheel for python-afl (pyproject.toml) ... done
  Created wheel for python-afl: filename=python_afl-0.7.3-cp310-cp310-linux_x86_64.whl size=164726 sha256=1a75ee45fd98edf7bb6f4322e45d601e20f43907dcf07bddb46fdb552f0197fc
  Stored in directory: /root/.cache/pip/wheels/dc/83/a7/2f416e489900e243aa272e4515a9360394ea5c33394132569b
Successfully built python-afl
Installing collected packages: python-afl
Successfully installed python-afl-0.7.3
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/war
ning/venv

```



```
[*] Baking aborted by the chef +++
[*] Writing output/default/fastresume.bin ...
[*] fastresume.bin successfully written with 279894 bytes.
[*] We're done here. Have a nice day!

[AF++ bdef6d55d05] /src # ls
output_main.py output
[AF++ bdef6d55d05] /src # cd output/
[AF++ bdef6d55d05] /src/output # ls
default
[AF++ bdef6d55d05] /src/output # cd default/
[AF++ bdef6d55d05] /src/output/default # ls
cdline fastresume bin fuzz bz libps fuzzer_setup fuzzer_stats hangs plot_data queue target_hash
[AF++ bdef6d55d05] /src/output/default # cd crashes
[AF++ bdef6d55d05] /src/output/default/crashes # ls
README.txt
ld:000000,sig:10,src:000000,time:1837,execs:750,op:havoc,rep:4 ld:000001,sig:10,src:000000,time:1926,execs:786,op:havoc,rep:3 ld:000003,sig:10,src:000000,time:19902,execs:7445,op:havoc,rep:3
[AF++ bdef6d55d05] /src/output/default/crashes # ls -l
ld:000002,sig:10,src:000000,time:11557,execs:4114,op:havoc,rep:4 ld:000004,sig:10,src:000010,time:39711,execs:15301,op:quick,pos:16
total 24
-rw----- 1 root root 553 Apr 1 13:43 README.txt
-rw----- 1 root root 4 Apr 1 13:43 ld:000000,sig:10,src:000000,time:1837,execs:750,op:havoc,rep:4
-rw----- 1 root root 4 Apr 1 13:43 ld:000001,sig:10,src:000000,time:1926,execs:786,op:havoc,rep:3
-rw----- 1 root root 4 Apr 1 13:43 ld:000002,sig:10,src:000000,time:11557,execs:4114,op:havoc,rep:4
-rw----- 1 root root 33 Apr 1 13:44 ld:000003,sig:10,src:000000,time:19902,execs:7445,op:havoc,rep:3
-rw----- 1 root root 22 Apr 1 13:44 ld:000004,sig:10,src:000010,time:39711,execs:15301,op:quick,pos:16
[AF++ bdef6d55d05] /src/output/default/crashes # cat ld:000000,sig:10,src:000000,time:1837,execs:750,op:havoc,rep:4
cat[AF++ bdef6d55d05] /src/output/default/crashes #
[AF++ bdef6d55d05] /src/output/default/crashes #
[AF++ bdef6d55d05] /src/output/default/crashes #
[AF++ bdef6d55d05] /src/output/default/crashes # cd ../hangs/
[AF++ bdef6d55d05] /src/output/default/hangs # ls -l
total 8
-rw----- 1 root root 22 Apr 1 13:43 ld:000000,sig:000000,time:4061,execs:1208,op:havoc,rep:3
-rw----- 1 root root 47 Apr 1 13:44 ld:000001,sig:000011,time:41933,execs:15880,op:havoc,rep:6
[AF++ bdef6d55d05] /src/output/default/hangs # ca
bash: ca: command not found
[AF++ bdef6d55d05] /src/output/default/hangs # cat id:000000,src:000000,time:4061,execs:1208,op:havoc,rep:3
test!!!!!!!!!!!!!![AF++ bdef6d55d05] /src/output/default/hangs # cat id:000000,src:000000,time:4061,execs:1208,op:havoc,rep:3
[AF++ bdef6d55d05] /src/output/default/hangs # cat id:000001,src:000011,time:41933,execs:15880,op:havoc,rep:6:15880,op:havoc,rep:6
!!!!!!!!!!!!!!!!!!!!id:000001,src:000011,time:41933,execs:15880,op:havoc,rep:6[AF++ bdef6d55d05] /src/output/default/hangs #
[AF++ bdef6d55d05] /src/output/default/hangs #
[AF++ bdef6d55d05] /src/output/default/hangs # cd ..
[AF++ bdef6d55d05] /src/output/default #
[AF++ bdef6d55d05] /src/output/default #
[AF++ bdef6d55d05] /src/output/default #
[AF++ bdef6d55d05] /src/output/default # cat fuzzer_stats
start_time      : 1743515020
last_update     : 1743515181
run time       : 160
fuzzer_pid     : 36
cycles_done    : 34
```

```
[AFL++ b0def0d55d05] /src/output/default/hangs # cd ../..
[AFL++ b0def0d55d05] /src/output/default #
[AFL++ b0def0d55d05] /src/output/default #
[AFL++ b0def0d55d05] /src/output/default # cat fuzzer_stats
start_time      : 1743515020
last_update     : 1743515181
run_time        : 160
fuzzer_pid      : 36
cycles_done     : 34
cycles_wo_finds : 32
time_wo_finds   : 120
fuzz_time       : 160
calibration_time : 0
cmplog_time     : 0
sync_time      : 0
trln_time       : 0
execs_done      : 73587
execs_per_sec   : 457.10
execs_ps_last_min : 485.16
corpus_count    : 14
corpus_favored  : 2
corpus_found    : 13
corpus_imported : 0
corpus_variable : 0
max_depth       : 5
cur_item        : 10
pending_favs    : 0
pending_total   : 0
stability       : 100.00%
bltnap_cvg      : 0.03%
saved_crashes   : 2
saved_hangs     : 2
total_tnout     : 381
last_find       : 1743515061
last_crash      : 1743515060
last_hang       : 1743515062
execs_since_crash : 58286
exec_timeout    : 20
slowest_exec_ms : 0
peak_rss_nb     : 14
cpu_affinity    : 0
edges_found     : 22
total_edges     : 65536
var_byte_count  : 0
havoc_expansion : 5
auto_dict_entries : 0
testcache_size  : 489
testcache_count : 14
testcache_evict : 0
afl_banner      : /usr/bin/python3
afl_version     : ++4.32a
target_mode     : shmem_testcase default
command_line    : afl-fuzz -i input -o output -- /usr/bin/python3 /src/main.py
[AFL++ b0def0d55d05] /src/output/default #
[AFL++ b0def0d55d05] /src/output/default #
[AFL++ b0def0d55d05] /src/output/default # cd ../..
[AFL++ b0def0d55d05] /src # ls
main.py
[AFL++ b0def0d55d05] /src # python3 main.py < output/default/crashes/id:000000,sig:10,src:000000,time:1837,execs:750,op:havoc,rep:4
Traceback (most recent call last):
  File "/src/main.py", line 23, in <module>
    print(uridecode(sys.stdin.read()))
  File "/src/main.py", line 11, in uridecode
    char_code = (int(a, 16) * 16) + int(b, 16)
ValueError: invalid literal for int() with base 16: 's'
[AFL++ b0def0d55d05] /src # timeout 5s python3 main.py < output/default/hangs/id:000001,src:000011,time:41933,execs:15880,op:havoc,rep:6
[AFL++ b0def0d55d05] /src #
```